This assignment will not be graded and is only for practice.

Level 1

1) Let $f(x, y)$ be defined in a neighbourhood of (a, b). When is (a, b) a critical point of f? ***1 point***

- [ ] $f(a, b) = 0.$
- [ ] $f_x(a, b) = 0.$
- [ ] $f_x(a, b) = 0 = f_y(a, b).$
- [ ] f_x does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(a, b) = 0 = f_y(a, b).$
f_x does not exist.

2) Consider $f(x, y) = \sqrt{5x^2 + 3y + 2}.$ ***1 point***

- [ ] Any point on the line $x + y = 1$ in the domain of definition of f satisfies $f_y = 0.$
- [ ] There is no (x, y) such that $f_y = 0.$
- [ ] Solutions of $5x^2 + 3y + 2 = 0$ are critical points of f.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There is no (x, y) such that $f_y = 0.$
Solutions of $5x^2 + 3y + 2 = 0$ are critical points of f.

3) A critical point ***1 point***

- [ ] is a point of maximum.

- [ ] may be a saddle point.
- [ ] may be a point of local minimum.
- [ ] can be obtained by equating f_x and f_y to 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

may be a saddle point.
may be a point of local minimum.
can be obtained by equating f_x and f_y to 0.

4) Let $f(x, y) = x^2y$. Then **1 point**

- [ ] $(0, 0)$ is a critical point.
- [ ] $(0, y)$ for $y \in \mathrm{R}$ are critical points.
- [ ] $f_x > 0$ at the critical points.
- [ ] $f_y = 0$ at the critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a critical point.
$(0, y)$ for $y \in \mathrm{R}$ are critical points.
$f_y = 0$ at the critical points.

5) For $f(x, y) = x^3 + 2xy - 2x - 4y$, which of the following is/are true? **1 point**

- [ ] $(2, -5)$ is a critical point.
- [ ] $(2, 5)$ is a critical point.
- [ ] $f_y = 0$ is a straight line parallel to the y-axis.
- [ ]

The system of equations $f_x = 0, f_y = 0$ has a unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(2, -5)$ is a critical point.
$f_y = 0$ is a straight line parallel to the y-axis.
The system of equations $f_x = 0, f_y = 0$ has a unique solution.

6) Choose the correct statement(s) about the function $f(x, y) = xy^2 + 2x$. ***1 point***

- [ ] $f_x = 0$ only for 2 values of x.
- [ ] $f_y = 0$ along both x and y axes.
- [ ] There are no critical points of f.
- [ ] The point $(0, 0)$ is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_y = 0$ along both x and y axes.
There are no critical points of f.

7) For $f(x, y) = \frac{x^3}{3} + y^2 + 2xy - 6x - 3y + 4$ ***1 point***

- [ ] $f_y = 0$ is a straight line passing through the origin.
- [ ] $(-1, \frac{5}{2})$ and $(3, -\frac{3}{2})$ are critical points.
- [ ] $f_x(-1, \frac{5}{2}) = 0$.
- [ ] $f_y(3, -\frac{3}{2}) = 2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-1, \frac{5}{2})$ and $(3, -\frac{3}{2})$ are critical points.
$f_x(-1, \frac{5}{2}) = 0$.

8) Choose the correct statement(s) about $f(x, y) = x^4 + y^4$. ***1 point***

- [ ] There are no critical points.
- [ ] There is exactly one critical point.
- [ ] $(0, 0)$ is a point of global minimum.
- [ ] $(0, 0)$ is a point of local minimum.

No, the answer is incorrect.
Score: 0

Accepted Answers:

There is exactly one critical point.
$(0, 0)$ is a point of global minimum.
$(0, 0)$ is a point of local minimum.

Level 2

9) Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$. Then **1 point**

- [ ] $(3, 1)$ is a critical point.
- [ ] $(2, 1)$ is a critical point.
- [ ] Value of f at $(2, 1)$ is -4.
- [ ] Value of f at $(3, 1)$ is -9.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(3, 1)$ is a critical point.
Value of f at $(2, 1)$ is -4.

10) Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$ be defined on the box $0 \leq x \leq 2$ and $0 \leq y \leq 1$. Then which of the following options is (are) true? **1 point**

- [ ] Along the line joining $(0, 0)$ and $(2, 0)$, the minimum is attained at $(2, 0)$.
- [ ] $(0, 1)$ is a point of extremum along the line joining $(0, 1)$ and $(2, 1)$.
- [ ] $f(0, 1) = 1$
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Along the line joining $(0, 0)$ and $(2, 0)$, the minimum is attained at $(2, 0)$.
$(0, 1)$ is a point of extremum along the line joining $(0, 1)$ and $(2, 1)$.

11) Let $f(x, y) = sin\ x\ cos\ y$. Then **1 point**

- [ ]

$((2k+1)\frac{\pi}{2}, k\pi)$ for $k \in \mathrm{N}$ are critical points of f.

- [ ] $((2k+1)\frac{\pi}{2}, (2k+1)\frac{\pi}{2})$ for $k \in \mathrm{N}$ are critical points of f.
- [ ] $(k\pi, k\pi)$ for $k \in \mathrm{N}$ are critical points of f.
- [ ] $(k\pi, (2k+1)\frac{\pi}{2})$ for $k \in \mathrm{N}$ are critical points of f.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$((2k+1)\frac{\pi}{2}, k\pi)$ for $k \in \mathrm{N}$ are critical points of f.
$(k\pi, (2k+1)\frac{\pi}{2})$ for $k \in \mathrm{N}$ are critical points of f.

12) Let $f(x,y) = e^{-x^2+y-\frac{y^3}{3}}$. Then **_1 point_**

- [ ] $f_x(1,0) = -\frac{2}{e}$.
- [ ] $f_x(0,0) = 1$.
- [ ] $f_y(0,-1) = 0$.
- [ ] $f_y(1,0) = e$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(1,0) = -\frac{2}{e}$.
$f_y(0,-1) = 0$.

13) Let $f(x,y) = e^{-x^2+y-\frac{y^3}{3}}$. Then **_1 point_**

- [ ] $(0,1)$ is a critical point.
- [ ] $(0,-1)$ is a critical point.
- [ ] $(1,1)$ is a critical point.

- [ ] $(1, -1)$ is a critical point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 1)$ is a critical point.
$(0, -1)$ is a critical point.

14) Find three positive numbers whose sum is 12 such that the sum of their squares is minimum. The mathematical expression that best defines this problem is ***1 point***

- ○ minimize $x^2 + y^2 + z^2$ such that $x + y + z = 12, x, y, z > 0.$
- ○ minimize $x + y + z$ such that $x^2 + y^2 + z^2 = 12, x, y, z \geq 0.$
- ○ minimize $x^2 + y^2 + z^2$ such that $x + y + z = 12, x, y, z \geq 0.$
- ○ minimize $x + y + z$ such that $x^2 + y^2 + z^2 = 12, x, y, z \geq 0.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

minimize $x^2 + y^2 + z^2$ such that $x + y + z = 12, x, y, z > 0.$

Your score is: 0/14